

GARV ARORA

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Robotics & Systems Engineer | Autonomous Systems • ROS2 • C/C++ • Python • Embedded RTOS • Computer Vision • AI Infrastructure

EDUCATION

Vellore Institute of Technology (VIT) — Vellore, India | Aug 2023 – July 2027

Bachelor of Technology in Computer Science & Engineering (Specialization: IoT) | CGPA: 8.36 / 10.0

PATENTS & KEY HONORS

- **Patent Published (App No: IN202641072249 A1):** *Autonomous Radar-Guided Survivor Detection & Navigation System* — Fused 24GHz FMCW mmWave radar, ultrasonic array, and IMU dead-reckoning on dual-core ESP32; formulated dynamic confidence grid decay $C = \max(0, C - \beta\Delta t)$ to detect breathing signatures under rubble without SLAM/LiDAR. (June 2026)
- **Amazon ML Summer School 2026:** Selected among Top 3,000 students out of 1,30,000+ applicants across India (~2.3% acceptance rate).
- **International Rover Challenge (IRC):** Ranked **13th Globally (2025)** and **17th Globally (2026)** with Team Vyadh (SEDS VIT).
- **TEDx Speaker:** Delivered talk on "Gender Stereotyping — Does It Still Exist?" at TEDxGEMSIInternationalSchool.

WORK EXPERIENCE

SEDS India (SEDS VIT) — *Research & Development Lead* | Jan 2026 – Present

- Promoted to R&D; Lead directing 2 national space robotics competition teams (50+ engineers) across Team Vyadh (IRC) and Team Ardra (ISDC); conducted architecture reviews and hardware debugging, resolving 10+ critical issues and securing 13th & 17th Global Ranks.

SEDS India (SEDS VIT) — *Autonomous Systems Developer* | Apr 2024 – Jan 2026

- Architected ROS2 Nav2 & RTAB-Map 3D Visual SLAM pipeline on Intel RealSense D455, generating real-time elevation maps and slashing manual teleoperation time by 70% during autonomous competition traversals.
- Slashed CV pipeline latency by ~60% and boosted throughput 4x using CUDA GPU kernels, YOLOv8, ArUco/AprilTag tracking, and Linux V4L2 zero-copy buffer tuning across 4 simultaneous camera streams.
- Flashed Micro-ROS firmware onto ESP32 microcontrollers for distributed sensor polling, fusing wheel encoders and 6-DOF IMU with an Extended Kalman Filter (EKF) to cut odometry drift by ~40% (sub-10 cm positioning accuracy).
- Shipped PyQt5 mission dashboard with real-time telemetry visualization, parameter tuning, and FSM mission execution.

Samsung PRISM — *Data Intelligence Agent Developer (Internship)* | Aug 2025 – Present

- Developed a multi-agent LangGraph pipeline (FastAPI, Redis, PostgreSQL, Qdrant) automating discovery, screening, and 20-stage analysis of HuggingFace/Kaggle dataset corpora for LLM training optimization.
- Integrated spaCy structural NLP, Presidio PII anonymization, SPDX license policy, BERTopic clustering, and MinHash LSH deduplication; implemented a MemorySaver Human-in-the-Loop review pause state.

LG Soft India (LGSi) — *HS/ES Lab* — *Living Solution Control Development Intern* | June 2026 – June 2026

- Engineered C/C++ embedded control logic and state machine handlers for living solution control modules, verifying execution timing on target hardware; work certified as "Technically Competent" by LGSi HR/Engineering management.

Wissen Baum Engineering Solutions LLP — *Software Automation Intern* | May 2025 – July 2025

- Built Python BDD test automation framework using Gherkin/Behave, pytest, Playwright, and Cypress, parallelizing GitLab CI jobs to cut pipeline execution time from ~10m to ~6m (40% speedup) and reducing manual testing effort by 80%+.

SELECTED PROJECTS

- **HayaiOS — Preemptive Bare-Metal RTOS Kernel** | C, ARM Assembly, ARM Cortex-M4
Implemented preemptive scheduler (1 ms tick) with context switching in <50 ARM assembly instructions, mutex/semaphore IPC primitives, and a hardware abstraction layer (HAL) isolating register-level I/O on ARM Cortex-M4.
- **Captivity CLI** | Rust, Python, Linux D-Bus, systemd, Socket IPC | [PyPI: captivity-cli](#)
Built autonomous captive portal WiFi login daemon featuring <50ms HTTP 204 probing, systemd background integration, D-Bus NetworkManager monitoring, keyring credential encryption, and dual Python-Rust TCP socket IPC.
- **3D Reconstruction — IMU-Enhanced KinectFusion** | C++, OpenCV, Intel RealSense SDK
Stabilized KinectFusion TSDF reconstruction by fusing RealSense D455 depth and 6-DOF IMU data via complementary filter, reducing tracking failures by ~50% across fast-motion trajectories.
- **teleop-cursor** | ROS 2, Python, rclpy, geometry_msgs/Twist | [PyPI: teleop-cursor](#)
Developed zero-hardware mouse cursor teleoperation node for ROS 2, converting screen displacement vectors into smooth linear/angular velocity commands (*geometry_msgs/Twist*) at 10 Hz callback frequency for TurtleBot3/Gazebo simulation.

TECHNICAL SKILLS

- **Languages:** C, C++, Python, Rust, ARM Assembly, Java, JavaScript, SQL, Bash, Verilog
- **Robotics & Autonomous Systems:** ROS2 (Nav2, MoveIt2), RTAB-Map (3D SLAM), Micro-ROS, EKF/UKF, PID & MPC Control, Sensor Fusion, mmWave Radar, Point Cloud Processing
- **Perception & Machine Learning:** OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, HuggingFace Transformers
- **Embedded & Infrastructure:** ARM Cortex-M, ESP32, FreeRTOS, I2C/SPI/UART, Bare-Metal C/Assembly, Linux Kernel/D-Bus, Docker, Git, GitLab CI/CD, AWS
- **Backend & AI Architecture:** LangGraph, FastAPI, Redis, PostgreSQL, Qdrant, REST APIs